

Pooled data and first differences

Slide route: Lecture10v2.tex: Starting with Pooled Data through Limits of First-Differences. Stop before Fixed Effects Models.

Session 13 | Same units or new samples?

Outcomes: Distinguish pooled cross-sections from panels, interpret time interactions, and show how first differences remove fixed unit characteristics.

100-minute route

0–10	Retrieve interaction interpretation.
10–30	Cross-section, pooled cross-section, fertility/gender examples.
30–45	Time interactions and changing associations.
45–55	Pause and identify the sampling unit.
55–75	Panel structure and traffic-fatality examples.
75–90	First differences for two and more periods.
90–100	Limits and exit exercise.

Slide-by-slide script

Slide 1 | Title

Guide Open by asking whether observations from two dates are the same people. That determines the data structure.

Slide 2 | Roadmap

Guide Preview cross-sections, pooled data, interactions over time, panels, and first differences.

Slide 3 | Last lesson

Guide Retrieve the IV distinction between a new source of variation and a new estimator.

Slide 4 | Starting with pooled data

Guide Name the unit, time index, outcome, and sample before interpreting any pooled regression.

Slide 5 | Cross-section

Guide Explain that one cross-section observes many units at one time.

Slide 6 | Pooled cross-section

Guide Contrast repeated independent samples with repeated observations of the same people.

Slide 7 | Pooled cross-section

Guide Use a time dummy to absorb a common average shift, but warn that it does not create a panel.

Slide 8 | Pooled cross-section

Guide Ask whether the composition of the sample can change over time and why that matters.

Slide 9 | Fertility over time

Guide Identify the outcome, cohort or time variable, and the reference category.

Slide 10 | Fertility over time

Guide Interpret a time indicator as a conditional difference, not automatically an individual change.

Slide 11 | Fertility over time

Guide Ask what omitted cohort characteristics could move with time.

Slide 12 | Fertility over time

Guide Use the interaction to let a relationship differ across periods.

Slide 13 | Fertility over time

Guide Calculate one fitted value from the displayed specification and identify the baseline group.

Slide 14 | Fertility over time

Guide Distinguish a time trend from a causal treatment effect.

Slide 15 | Gender gap

Guide Write the group contrast at the reference date and explain its units.

Slide 16 | Gender gap

Guide Use the interaction to calculate how the gap changes later.

Slide 17 | Gender gap

Guide Ask whether the later gap difference is causal without a parallel-trends argument.

Slide 18 | Testing differences

Guide Frame the changing-gap question as a joint or linear restriction when appropriate.

Slide 19 | Transition to panel data

Guide Explain why following the same units changes what can be removed.

Slide 20 | Panel data

Guide Define a panel using unit and time indices. Have students label both in the traffic-fatality example.

Slide 21 | Panel data

Guide Contrast between-unit variation with within-unit changes.

Slide 22 | Why panel data?

Guide Explain that fixed characteristics can be differenced out, while time-varying confounders remain.

Slide 23 | Traffic fatalities, 1982

Guide Read one state's outcome and beer tax, then identify the comparison being made.

Slide 24 | Traffic fatalities, 1983

Guide Ask students to compute the change for one state.

Slide 25 | Traffic fatalities, 1984

Guide Continue the table and distinguish levels from changes.

Slide 26 | Traffic fatalities, 1985

Guide Ask whether a common time shock would disappear from first differences.

Slide 27 | Traffic fatalities, 1986

Guide Keep state labels fixed; this is a panel, not a new cross-section.

Slide 28 | Traffic fatalities, 1987

Guide Use one state as a numerical check on the differencing arithmetic.

Slide 29 | Traffic fatalities, 1988

Guide Ask what information is lost when unit means are removed.

Slide 30 | First differences

Guide Write $\Delta Y_{it} = \beta \Delta X_{it} + \Delta u_{it}$ and show the fixed effect cancels.

Slide 31 | First differences

Guide Use the three-unit example from this guide to calculate the slope through the origin with a common change.

Slide 32 | First differences when $T > 2$

Guide Explain that successive differences share level errors and can be serially correlated.

Slide 33 | Limits of first differences

Guide List time-varying confounding, measurement error amplification, and the need for repeated observations.

Slide 34 | Limits of first differences

Guide Ask whether independent repeated cross-sections can be differenced person by person. No.

Slide 35 | Transition

Guide Close by previewing within transformations and fixed effects for the next class.

Side example | Pooled observations

Use with slides Lecture10v2 slides 5–18: Cross-Section, Pooled Cross-Section, Fertility over time, Gender Gap, and Testing Differences in Pooled regressions. Use the two-by-two interaction example while moving through slides 16–18.

In a synthetic pooled wage model,

$$wage = 10 + 2D_{\text{later}} + 3D_{\text{group}} - 1(D_{\text{later}}D_{\text{group}}) + u,$$

the group gap is 3 initially and 2 later. The interaction is the change in the gap, -1 . With new people sampled at each date, this is not the average change in the same individuals' wages.

Ask Does a two-by-two interaction automatically identify a causal difference-in-differences effect? No. That reading requires a design and additional assumptions such as a suitable parallel-trends counterfactual. Here the example is an interpretation exercise.

Panel definition A panel repeatedly observes the same units. Time labels on independent cross-sections do not turn them into a panel. Have students name the unit, time index, and outcome before interpreting any coefficient.

Worked example | Removing a fixed effect

Use with slides Lecture10v2 slides 20–34: panel data through Limits of First-Differences. Use the three-unit table when the deck reaches Traffic Fatalities first differences, especially slides 31–34.

Suppose

$$Y_{it} = \alpha_i + \beta X_{it} + \lambda_t + u_{it}.$$

With two periods,

$$\Delta Y_i = \beta \Delta X_i + \Delta \lambda + \Delta u_i.$$

The unit effect α_i cancels. A common time shock remains and can be captured by an intercept in the two-period differenced regression.

Unit	X_1	X_2	Y_1	Y_2
A	1	2	10	13
B	2	4	20	25
C	3	6	30	37

Differences are $(\Delta X, \Delta Y) = (1, 3), (2, 5), (3, 7)$. The fitted differenced relationship is

$$\Delta \hat{Y} = 1 + 2\Delta X.$$

Ask What would happen if we forced this line through the origin? The slope would be $(3 + 10 + 21)/(1 + 4 + 9) = 34/14 \simeq 2.429$. Omitting the common change contaminates this synthetic comparison.

Identification comment Removing fixed attributes does not remove time-varying confounders. If a district changes tax policy in response to an unobserved changing risk, first differences need not identify the causal effect.

More than two periods Successive differences share errors: $\Delta u_t = u_t - u_{t-1}$ and $\Delta u_{t-1} = u_{t-1} - u_{t-2}$. Even independent level errors induce dependence between neighboring differenced errors.

Exit answers Time-invariant regressors disappear under differencing. A common time shock does not disappear unless it is unchanged. Repeated independent samples cannot generally be differenced person by person.

If short / preparation Show selected years of the traffic-fatality figures rather than each in detail. Preserve the three-unit example. Next time extend removal of fixed attributes to all periods using within transformations.

Source: Lecture10v2.tex, pooled-data and first-difference sections; synthetic numerical table.